

Data dictionary

One record per customer. Survey characteristics are measured at the start of the campaign; all history variables are measured before treatment assignment. 30d means 30 days and 90d means 90 days. These baseline windows end immediately before assignment. The campaign outcome window is the following 30 days.

Variable	Meaning	Values / units
CustomerID	Artificial customer identifier	HG00001-HG10000; not a quantitative predictor
Age	Customer's age at assignment	Whole years, 18-80
HouseholdIncome	Annual household income reported in the panel survey	Dollars; blank if missing
HouseholdSize	People living in the household, including the customer	Integer, 1-6
ChildrenAtHome	Children living in the household	Integer, 0 through HouseholdSize minus 1
Homeowner	Household owns its primary residence	1 = owns; 0 = does not own
Region	US region of residence	Northeast, Midwest, South, West
Education	Customer's highest completed education	High school or less; Some college/associate; Bachelor's; Graduate
HomeSizeSqFt	Floor area of primary residence from the survey	Square feet; blank if missing
SiteVisits30d	Website sessions in the 30 days before assignment	Nonnegative integer
ProductViews30d	Product-page views in the same prior 30 days	Nonnegative integer; repeated views count
DaysSinceLastVisit	Days since the most recent website session at assignment	Whole days; 0 means less than one day; over 30 when SiteVisits30d = 0
EmailEngagement90d	Fraction of prior marketing emails with at least one click in the preceding 90 days	0-1; all panel members received at least one prior email
PreviousSpend	Total spending at this retailer in the 12 months before assignment	Dollars
Treatment	Random assignment to the campaign email	1 = assigned email; 0 = no campaign email
Conversion	Made at least one purchase in the 30 days after assignment	1 = purchased; 0 = did not purchase
CampaignSpend	Total spending at this retailer in those same 30 days	Dollars; 0 for nonconverters; positive for converters

Treatment records assignment, not whether the customer opened or clicked the new email. EmailEngagement90d refers only to **earlier** messages. Conversion and CampaignSpend are measured after assignment. PreviousSpend is historical and does not include CampaignSpend.